COMSATS University Islamabad (CUI)

Project Proposal

for

AI-Assisted Plasma Process Experimentation & Digital Twin Analytics Platform

Scope Document — Version 3.1

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October 2026

Revision History

The table below records every released version of this scope document.

Version	Date	Description of change	Author(s)
3.0	25-Oct-2026	First scope document of the project: a three-stage plasma digital twin (low-pressure argon plasma with optical emission spectroscopy; underwater oxygen plasma for disinfection; Langmuir probe diagnostics), defined from the supervisor's written questionnaire answers.	M. Rayyan Taimoor Khan, Sadeem Ur Rehman
3.1	29-Oct-2026	AI made the second core module (no longer optional): the project now has Core Module 1, the Digital Twin, and Core Module 2, the AI Module (spectrum inference, physics-anomaly detection, surrogate with SHAP, twin-verified recommendations, training and registry). Objectives, modules, tools, work division and timeline updated to match SRS v1.1.	M. Rayyan Taimoor Khan, Sadeem Ur Rehman

Abstract

Rationale: Plasma physics students need to practise optical emission spectroscopy (OES) and Langmuir probe diagnostics, but laboratory plasma systems are expensive, shared and slow to operate, so hands-on time is scarce.

Existing Systems: Commercial multiphysics simulators such as the COMSOL Plasma Module are costly and need finite-element expertise, while free tools such as BOLSIG+, SPECAIR, MassiveOES and vendor probe software each cover only one step (electron kinetics, spectrum synthesis or curve fitting). None of them connects discharge settings to a solved plasma state, to the resulting diagnostic signals and to their validation.

Targeted Challenges: A student cannot see how a change in RF power, pressure or gas flow propagates through the plasma into the measured spectrum and probe curve, and there is no simple, tolerance-based way to check two diagnostics against each other or against measured data. For underwater plasma disinfection, discharge settings are rarely linked quantitatively to the reactive species produced and the resulting bacterial log reduction.

Grey Area: No low-cost, physics-based tool links discharge settings, plasma state, synthetic diagnostics and validation for both a low-pressure plasma and an underwater disinfection plasma in one environment.

Objectives: The platform will simulate a 13.56 MHz capacitively coupled argon plasma (10–100 W, 0.1–0.5 mbar, Ar 100–500 sccm, with optional N_2 and O_2) whose electron temperature, electron density and gas temperature come from particle and power balance equations. It will generate and analyse virtual OES spectra and virtual Langmuir probe curves for that plasma and compare them with the model within a fixed ±30 % tolerance. It will also model an underwater pulsed-DC oxygen bubble discharge whose reactive species drive a Hom-model prediction of *E. coli* inactivation, with validation against published and measured data. A second core module applies AI to the twin: it infers plasma parameters from spectra with stated uncertainty, detects runs that break the expected physics, explains the twin through a fast surrogate, and recommends settings that are verified by the twin.

Significance: The result is a free, repeatable virtual plasma laboratory in which students and researchers can rehearse experiments, interpret diagnostics and validate measurements before and after using the real laboratory.

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1. Introduction

Plasma diagnostics are how physicists learn what a plasma is doing. Optical emission spectroscopy reads the light emitted by excited atoms and molecules to infer temperatures and densities, and a Langmuir probe, a small electrode inserted into the plasma, measures a current–voltage curve from which the electron temperature, the electron density and the electron energy distribution are extracted. Learning to use and interpret these techniques requires many repeated experiments, which are hard to obtain on shared laboratory equipment.

This project builds a software digital twin of the experiments defined by the supervisor, a plasma physicist whose research idea the project is. It has three stages. **Stage 1** is a low-pressure argon plasma generated in a 13.56 MHz capacitively coupled (CCP) chamber and diagnosed by OES. **Stage 2** is an underwater, high-pressure oxygen plasma generated inside gas bubbles by a pulsed DC supply and used to disinfect municipal waste water; OES is applied to it as well. **Stage 3** is Langmuir probe diagnostics, applied to the Stage 1 plasma only. Stage 1 is the most important stage, but all three are mandatory.

For each stage, the user sets the controls that exist on the real equipment, the platform solves physics-based models for the plasma state, and it then produces the signal a real instrument would record: an emission spectrum, a probe current–voltage curve or a bacterial survival curve. These signals are analysed with the same methods used in the laboratory, compared with each other and with the model, and validated against published data and, when available, against measurements from the laboratory. The platform has two core modules: **Core Module 1, the Digital Twin**, which holds all of this physics, and **Core Module 2, the AI Module**, which learns from data generated by the twin to infer plasma parameters from spectra, detect physics-relationship anomalies, explain the twin through a fast surrogate, and recommend settings that are verified by re-running the twin.

The platform is developed in Python with Streamlit for the user interface, Plotly for interactive charts, NumPy and SciPy for the numerical models, scikit-learn, XGBoost and SHAP for the AI module, and SQLite for storing sessions. It runs on an ordinary laptop with no laboratory hardware, and it demonstrates how software engineering, numerical computing, machine learning, databases and human–computer interaction combine to model a real physical process.

2. Problem Statement

Plasma diagnostics are learned by doing, but doing is expensive. A low-pressure RF plasma system needs a vacuum chamber, pumps, gas controllers, an RF generator with a matching network, a spectrometer and a probe system, and the laboratory is shared between many users. A student therefore gets only a few hours of operating time, often without being able to repeat a run or explore the full range of power, pressure and gas flow.

Interpreting the results is also hard. The emission spectrum depends on the electron temperature, the electron density and the electron energy distribution through a chain of atomic processes, and a probe curve is distorted by the RF sheath oscillation. Students who see only the final spectrum or curve cannot tell which physical cause produced which feature, and they rarely check one diagnostic against another.

Existing software does not close this gap. General multiphysics packages are costly and require expertise in finite-element modelling, while free tools each address one step and must be joined by hand in spreadsheets. The same problem exists for underwater plasma disinfection, where the link from pulse voltage, frequency and gas flow to reactive species and finally to bacterial log reduction is usually explored only by trial and error.

Finally, when measured data become available, there is no consistent, tolerance-based way to compare them with theory and with the other diagnostic, so errors in either the experiment or the analysis can go unnoticed. This project addresses all of these problems with one integrated, physics-based virtual laboratory.

3. Problem Solution / Objectives of the Proposed System

The proposed system provides a digital twin for each plasma and a virtual instrument for each diagnostic, all connected through one solved plasma state. For Stage 1, a volume-averaged (0D) global model solves the particle balance for the electron temperature and the power balance for the electron density [1, 2], extended with gas flow, gas temperature, $Ar/N_2/O_2$ chemistry, a Druyvesteyn electron energy distribution and a 1D axial profile. The virtual OES uses a collisional-radiative (CR) model of the argon 1s and 2p levels [3] and NIST line data [4]. The virtual Langmuir probe computes an RF-distorted current–voltage curve and analyses it with orbital-motion-limited (OML) theory and the Druyvesteyn method [5, 6]. For Stage 2, a gas-phase global model of the oxygen–water-vapour bubble discharge feeds an empirical transfer of reactive species into the water and a Hom disinfection model [7]. A comparison function checks every diagnostic against the model with a fixed ±30 % tolerance, and measured data are imported for validation only. The AI module learns from data generated by the twin: tree-ensemble models infer plasma parameters from a spectrum with conformal intervals, an Isolation Forest flags diagnostics that break the expected physics, a surrogate with SHAP explains the twin, and a recommendation engine proposes settings and re-simulates each one through the twin.

3.1 Primary Objectives

BO-1: Provide a virtual low-pressure argon CCP experiment that covers the whole supervisor-defined operating envelope (10–100 W, 0.1–0.5 mbar, Ar 100–500 sccm, N_2 and O_2 admixtures) and whose computed T_e, n_e and T_{gas} agree with published reference values within ±30 % at no fewer than three reference operating points.

BO-2: Allow a student to configure a Stage 1 run and obtain a synthetic OES spectrum and a Langmuir probe analysis for it in under 5 minutes, with every derived quantity (T_e, n_e, T_{gas}, T_{exc}) reported with its uncertainty and method.

BO-3: Demonstrate diagnostic consistency: OES- and probe-derived values shall agree with the model within ±30 % for at least 90 % of a pre-registered grid of Stage 1 test configurations.

BO-4: Predict the *E. coli* log reduction of the underwater oxygen discharge within ±30 % of published reference data at no fewer than one reference operating point, together with the D-value and energy yield.

BO-5: Allow a researcher to import a measured OES spectrum (.txt) or probe I-V curve (.csv) and obtain a validation report in under 10 minutes, without measured data ever being used to tune model parameters.

BO-6: Infer T_e, n_e and T_{gas} from an emission spectrum with an AI model whose error is significantly lower than a ridge-regression baseline (paired bootstrap, $p < 0.05$) and whose 90 % prediction intervals reach at least 85 % coverage on held-out synthetic spectra.

BO-7: Detect physics-relationship faults injected by the twin with at least 90 % recall at no more than 5 % false alarms, significantly better than a range-check baseline.

BO-8: Provide a surrogate of the twin with $R^2 \geq 0.98$ and median relative error ≤ 5 % on held-out twin runs, at least 100 times faster than the twin, with SHAP explanations of its outputs.

BO-9: Recommend operating settings for a user goal such that 100 % of recommendations are re-simulated by the twin and the twin confirms the goal is met for at least 90 % of a benchmark set of goals.

3.2 Secondary Objectives

BO-10: Deliver the complete platform as a standalone Python application that installs from one pinned requirements file and runs without laboratory hardware on standard university computers.

BO-11: Make every result reproducible and defensible: fixed random seeds, a model version stored with every session and AI model, and every constant and reference value stored with its published source.

4. Related System Analysis / Literature Review

Global discharge models. For a low-pressure discharge, the particle balance between volume ionization and wall loss fixes the electron temperature independently of power, and the power balance then fixes the electron density [1]. This approach has been extended to argon–oxygen

mixtures with negative ions [2], and published benchmarks for capacitive discharges provide reference values [8]. Global models are fast enough for interactive use, which is why they are chosen instead of fluid or particle-in-cell codes.

Electron energy distribution. At tens of pascals the distribution in an RF argon discharge departs from a Maxwellian [9]; the Druyvesteyn form is the supervisor's choice for low pressure. Rate coefficients are integrated from LXCat cross-sections [10] over the chosen distribution, as in the BOLSIG+ solver [11].

Optical emission spectroscopy. Collisional-radiative models of the argon 1s and 2p levels [3] and line-ratio methods for argon–nitrogen plasmas [12] show that the 2p levels are not in Boltzmann equilibrium, so an excitation temperature must not be reported as the electron temperature. N_2 rotational temperatures in argon-rich plasmas can be distorted by metastable energy transfer [13], which is why the (1,0) band at 315.9 nm with an NH overlap check is used. For the dense underwater plasma, H-alpha Stark broadening gives the electron density [14].

Langmuir probes and the local laboratory. OML theory and RF distortion are described by Chen [5]. A study from the COMSATS Plasma Research Laboratory used a single tungsten probe of radius 0.195 mm and 5 mm length swept from −20 to +50 V and obtained the EEDF by the Druyvesteyn method [15]; these values are the defaults.

Underwater plasma disinfection. Discharges in and in contact with water produce O_3, H_2O_2, OH and UV, which transfer into the liquid [16, 17]. A COMSATS-linked underwater discharge study reported OH, H-alpha and atomic oxygen lines and the reduction of *E. coli* [18]. Typical municipal waste-water properties are taken from Metcalf & Eddy [19].

Machine learning on simulation data. Tree ensembles, such as Random Forests and XGBoost [20, 21], outperform deep networks on tabular data of this kind [22]. Conformal prediction adds intervals of stated coverage [23], Isolation Forest finds anomalies without labelled faults [24], SHAP explains each prediction [25], and differential evolution searches a bounded design space without gradients [26]. Because training data come from the twin, every AI output can be checked against a known physical answer, and simple baselines show whether the learning adds value.

Table 4.1 compares existing systems with the proposed platform.

Table 4.1: Related system analysis with proposed project solution

Application / System	Weaknesses / Limitations	Proposed Platform Solution
COMSOL Multiphysics Plasma Module [27]	Expensive licence; needs finite-element expertise; no built-in synthetic OES or probe workflow; slow for teaching sweeps.	Free, interactive global model with integrated virtual OES and probe; runs in seconds on a laptop.
BOLSIG+ [11]	Computes the EEDF and rate coefficients only; no plasma state, spectrum or probe curve.	Uses the same cross-section approach inside a complete plasma-to-diagnostic chain.

Application / System	Weaknesses / Limitations	Proposed Platform Solution
SPECAIR [28]	Synthesises spectra from user-supplied temperatures; does not compute the plasma state.	Spectrum is computed from the solved plasma state through a CR model.
MassiveOES [29]	Fits temperatures to measured molecular spectra; no plasma model or probe.	Combines spectrum analysis with a forward model and a probe cross-check.
Impedans probe software [30]	Tied to vendor hardware; analyses measured curves only.	Generates RF-distorted synthetic curves and analyses synthetic and imported curves with open methods.
PlasmaPy [31]	Library of plasma formulas; no user interface or experiment workflow.	Guided point-and-click workflow with saved sessions and reports for students.

5. Vision Statement

For plasma physics students and researchers at COMSATS University Islamabad **who** need to practise and understand plasma diagnostics without waiting for scarce laboratory time, **the** Plasma Digital Twin Platform **is** a physics-based virtual plasma laboratory **that** lets them configure a low-pressure argon plasma and an underwater oxygen plasma, obtain the resulting plasma state, generate and analyse realistic OES spectra and Langmuir probe curves, predict disinfection performance, and validate every result against published and measured data. **Unlike** specialist commercial simulators and single-purpose analysis programs, which are expensive or cover only one step of the chain, **our product** connects discharge settings, plasma physics, synthetic diagnostics and validation in one free, transparent and repeatable environment in which every number shown can be traced back to a published equation or data source.

6. Scope

The scope of the project is a single-user, locally installed software platform that simulates and analyses two laboratory plasmas and two diagnostic techniques. The platform simulates a low-pressure argon plasma in a 13.56 MHz capacitively coupled chamber of 40 cm by 40 cm with electrodes 14 cm in diameter and 4.5 cm apart. The user controls RF power from 10 to 100 W, pressure from 0.1 to 0.5 mbar, argon flow from 100 to 500 sccm and the gas mixture, where nitrogen and oxygen are each added as 5 % of the argon flow by default. The Stage 1 model is a volume-averaged global model that solves particle and power balance for the electron temperature and electron density and an energy balance for the gas temperature, with a Druyvesteyn electron energy distribution and an optional one-dimensional axial profile. A virtual OES module produces the emission spectrum of this plasma from a collisional-radiative model, adds the effect of the spectrometer resolution and noise, and analyses the spectrum with a Boltzmann plot, the excitation temperature, the excited-level population distribution and the gas temperature from Doppler broadening or the nitrogen rotational band. A virtual Langmuir probe produces the current–voltage curve of a cylindrical tungsten probe in the same plasma, including RF sheath distortion, and analyses it with orbital-motion-limited theory to obtain the floating and

plasma potentials, electron temperature, electron density, Debye length and the electron energy distribution. A comparison function shows model, OES and probe values side by side and marks each as inside or outside a fixed ±30 % tolerance.

The platform also simulates an underwater oxygen plasma generated inside bubbles by a pulsed DC supply of up to 10 kV at 5 to 10 kHz and 100 W average power across a pin-to-pin gap of 1 to 3 mm at 1 atm. For this plasma it computes the bubble plasma state with a Maxwellian distribution, the concentrations of ozone, hydrogen peroxide and hydroxyl radicals in the water, the change in pH, oxidation–reduction potential, conductivity and temperature, and the survival of *E. coli* and other listed organisms with the Hom model. Its virtual OES shows the OH, H-alpha and atomic oxygen lines and estimates the electron density from H-alpha Stark broadening.

In researcher mode, the user can import measured spectra and probe curves, attach the required metadata, and validate the models against these files and against published benchmarks, while measured data stay non-citable unless permission is given. All runs can be saved, reopened, compared and exported as tables, figures or a PDF report.

The AI module is the second core module and is trained only on synthetic data generated by the twin. It infers electron temperature, electron density and gas temperature from a spectrum with prediction intervals; flags runs and measured files whose diagnostics break the expected physics relationship; offers a fast surrogate of the twin with SHAP explanations; and recommends settings for a user goal, each of which is re-simulated by the twin before it is shown. Every AI model is compared with a simple baseline, evaluated on data it has not seen, and logged with its training data, seed and the twin version it learned from.

The scope explicitly excludes control of, or data acquisition from, real laboratory hardware; feed gases other than argon with nitrogen and oxygen admixtures; two- and three-dimensional fluid or particle-in-cell simulation; full plasma–liquid reaction–transport modelling; multi-user accounts and cloud deployment; and certified engineering accuracy beyond the stated ±30 % semi-quantitative tolerance.

7. System Modules

The platform has **two core modules** and two supporting components. Core Module 1, the Digital Twin, holds all the physics: sub-modules 1.1 and 1.2 implement the supervisor's Stage 1, sub-module 1.4 implements Stage 2 and sub-module 1.3 implements Stage 3 (the Langmuir probe, applied to the Stage 1 plasma only). Core Module 2, the AI Module, learns from data generated by the twin and always calls the twin to verify what it proposes; the twin never depends on the AI, so the physics can always be tested and defended on its own. The twin is built first, because the AI learns from it.

7.1 Core Module 1: Digital Twin (physics engine)

Sub-module 1.1 Low-Pressure Plasma Model (Stage 1)

The core physics engine. Electron temperature comes from the particle balance by root finding, electron density from the power balance, and every other output is derived from these two values.

FE-1.1.1: Configure RF power, pressure, argon flow, gas mixture, chamber geometry and model options within the supervisor envelope.

FE-1.1.2: Solve the 0D global model: particle balance for T_e and power balance for n_e.

FE-1.1.3: Compute the gas temperature from an energy balance and the species densities for Ar, N_2 and O_2 mixtures, including metastable and stepwise processes.

FE-1.1.4: Use a Druyvesteyn (default) or Maxwellian EEDF with rate coefficients integrated from published cross-sections.

FE-1.1.5: Compute 1D axial profiles between the electrodes.

FE-1.1.6: Run parameter sweeps of up to 200 points.

Sub-module 1.2 Virtual Optical Emission Spectroscopy

Turns the solved plasma state into the spectrum a spectrometer would record, then analyses it as in the laboratory.

FE-1.2.1: Compute excited-level populations with a collisional-radiative model.

FE-1.2.2: Synthesise an emission spectrum with instrument resolution, wavelength range and noise.

FE-1.2.3: Build a Boltzmann plot and report the excitation temperature.

FE-1.2.4: Display excited-level populations against the Boltzmann distribution.

FE-1.2.5: Determine the gas temperature by N_2 rotational band or Doppler broadening.

FE-1.2.6: Estimate T_e by CR-model line ratios and analyse imported measured spectra.

Sub-module 1.3 Virtual Langmuir Probe and Diagnostics Comparison (Stage 3)

Generates and analyses the probe curve of the same plasma, giving an independent check on the OES and the model.

FE-1.3.1: Generate a synthetic I-V curve for a cylindrical tungsten probe with RF sheath distortion and noise.

FE-1.3.2: Analyse the curve with OML theory for V_f, V_p, T_e, n_e and λ_D, including an OML validity check.

FE-1.3.3: Extract the EEPF with the Druyvesteyn second-derivative method.

FE-1.3.4: Compare model, OES and probe values within the ±30 % tolerance.

Sub-module 1.4 Underwater Plasma Disinfection Model (Stage 2)

Links pulsed-discharge settings to reactive species in the water and to bacterial survival.

FE-1.4.1: Configure the pulsed DC supply, electrode gap, oxygen flow, bubble gas, water properties and target organism.

FE-1.4.2: Check pulse consistency (duty cycle and energy per pulse).

FE-1.4.3: Solve the bubble gas-phase global model with a Maxwellian EEDF.

FE-1.4.4: Transfer O_3, H_2O_2 and OH into the liquid with empirical coefficients and compute pH, ORP, conductivity and temperature change.

FE-1.4.5: Predict survival with the Hom model and report log reduction, D-value and energy yield.

FE-1.4.6: Synthesise the underwater OES and estimate n_e from H-alpha Stark broadening.

Sub-module 1.5 Validation and Measured Data (Researcher)

Brings laboratory measurements in for validation only; measured data never tune the models.

FE-1.5.1: Import measured OES spectra (.txt) and probe I-V curves (.csv) with column mapping and units.

FE-1.5.2: Record mandatory metadata and the citation permission for each file.

FE-1.5.3: Run literature benchmarks and measured-data comparisons with the fixed ±30 % tolerance.

FE-1.5.4: Generate a validation report.

7.2 Core Module 2: AI Module

Sub-module 2.1 Spectrum-to-Parameter Inference

Reads plasma parameters directly from an emission spectrum, with honest uncertainty.

FE-2.1.1: Infer T_e, n_e and T_{gas} from a synthetic or imported spectrum with Random Forest or XGBoost regressors.

FE-2.1.2: Give 90 % conformal prediction intervals and withhold estimates outside the training range.

FE-2.1.3: Show the physics-based analysis next to every AI estimate.

FE-2.1.4: Compare every model with a ridge-regression baseline using a paired bootstrap test.

Sub-module 2.2 Physics-Relationship Anomaly Detection

Finds diagnostics that disagree with the physics, which a simple range check cannot see.

FE-2.2.1: Compute residuals between the twin's prediction and the observed diagnostics of a run or measured file.

FE-2.2.2: Score the residuals with an Isolation Forest and assign Normal, Warning or Critical severity.

FE-2.2.3: Name the residual that contributed most as the likely cause.

FE-2.2.4: Report detection quality on held-out injected faults against a range-check baseline.

Sub-module 2.3 Surrogate Model and Explainability

A fast learned copy of the twin for large explorations, explained input by input.

FE-2.3.1: Train fast surrogates of the Stage 1 and Stage 2 twins for large explorations.

FE-2.3.2: Explain outputs with global and local SHAP values.

FE-2.3.3: Check any surrogate value with the twin and show the twin value if they disagree.

Sub-module 2.4 Counterfactual Recommendation (Inverse Design)

Turns a goal into settings, and proves each proposal by re-running the twin.

FE-2.4.1: Accept a goal (for example, a target log reduction with the least energy) and fixed settings.

FE-2.4.2: Search the operating envelope on the surrogate and re-simulate the best candidates with the twin.

FE-2.4.3: Rank candidates by their twin-verified outcome and report what-if counterfactuals.

FE-2.4.4: State clearly when no setting inside the envelope can reach the goal.

Sub-module 2.5 AI Training, Evaluation and Model Registry

Keeps the AI honest and reproducible.

FE-2.5.1: Generate training data with the twin, with noise, faults and seed fixed before training.

FE-2.5.2: Evaluate every model on random and region-based (extrapolation) splits.

FE-2.5.3: Log every training run with MLflow and publish a model card.

FE-2.5.4: Stamp each model with the twin version and flag models trained on an older version.

7.3 Supporting Components

Sub-module S.1 Home, Configuration and Session Management

Gives one entry point to all stages and stores every run so it can be reopened and compared.

FE-S.1.1: Select the usage role (Student or Researcher) without login.

FE-S.1.2: Start a default experiment in one action from the quick-start list.

FE-S.1.3: Save, list, search, reopen, compare and delete sessions (configuration plus results).

FE-S.1.4: Show the load status of every reference data set and AI model.

Sub-module S.2 Visualisation, Reporting and Export

Presents every result clearly and exports it for reports.

FE-S.2.1: Show every result as an interactive chart with labelled axes and SI units.

FE-S.2.2: Export results tables, figures and raw spectra or I-V data as PDF, CSV or PNG.

FE-S.2.3: Exclude non-citable measured data from every export.

FE-S.2.4: Provide a user guide and a glossary of plasma terms.

8. System Limitations / Constraints

LI-1: The platform is software-only. It does not control or acquire data directly from laboratory hardware; measured data enter only as exported files.

LI-2: Stage 1 is limited to argon with nitrogen and/or oxygen admixtures of 1–10 % of the argon flow each; other feed gases are not modelled.

LI-3: Stage 1 uses a volume-averaged (0D) model with an optional 1D axial profile; radial non-uniformity, 2D/3D fluid and particle-in-cell effects are not modelled.

LI-4: When oxygen is present the plasma becomes electronegative; the global model includes negative ions, but the OML probe analysis assumes an electropositive plasma, so probe results with O_2 are flagged as approximate.

LI-5: Stage 2 uses a gas-phase global model with empirical transfer of species into the liquid. It does not solve full plasma–liquid reaction–transport equations or bubble dynamics.

LI-6: Hom disinfection parameters come from published time-resolved data until validated measurements exist. An organism without published parameters cannot be simulated and is shown as unavailable.

LI-7: Measured data supplied by the laboratory may not be cited or published (supervisor decision); report validation therefore relies on published literature, and measured data are used privately for validation only.

LI-8: Agreement is targeted at ±30 % (semi-quantitative). The platform is a learning and pre-experiment planning tool, not a certified predictive instrument.

LI-9: The platform is single-user and runs on one computer; there are no user accounts, so the Student and Researcher roles are usage modes, not security boundaries.

9. Data Gathering Approach

- **Supervisor questionnaires:** the operating envelopes, diagnostic methods, organisms, tolerance and data rules were collected from the supervisor through written questionnaires; his answers are authoritative, and every value is recorded with its source.

- **Literature review:** plasma physics, spectroscopy, probe and disinfection literature (IEEE Xplore, ScienceDirect, IOP, arXiv) supplies the models, the reference benchmarks and any value the supervisor left blank.
- **Local laboratory publications:** published work from the COMSATS Plasma Research Laboratory provides chamber, probe and spectrometer details so that defaults match the real set-up.
- **Published reference data:** NIST atomic line data, LXCat cross-sections, N_2 band constants and published survival data are installed with the application and cited.
- **Twin-generated training data for the AI module:** datasets produced by running the digital twin across the operating envelope, with noise levels, injected faults and seed fixed before any model is trained.
- **Measured laboratory data (planned):** the supervisor will provide sample OES text files and probe .csv files. They are held out for validation only and are not cited.
- **Prototype walkthroughs and usability feedback:** a clickable mockup prototype is reviewed with the supervisor and fellow students to refine the workflow before and during development.

10. Project Contribution

- **One plasma state for every diagnostic:** the spectrum, the probe curve and the comparison are all computed from the same solved plasma state, so any feature in a diagnostic signal can be traced to its physical cause.
- **Physics instead of curve fitting:** electron temperature and density come from particle and power balance equations, and every other output is derived from them.
- **Built-in cross-validation:** OES, probe and model values are compared automatically against a tolerance fixed before the comparison.
- **Honest treatment of non-equilibrium:** the excitation temperature and CR-model populations are shown together with an explanation of why the excitation temperature is not the electron temperature.
- **Two plasmas in one tool:** the same diagnostic framework covers a low-pressure argon plasma and an underwater disinfection plasma, linking discharge settings to bacterial log reduction and energy yield.
- **Responsible use of measured data:** measured files are used for validation only and non-citable data are removed from every export automatically.
- **AI that learns from physics and is checked by it:** the AI module is trained only on twin data, every AI estimate sits beside the physics result, and every recommendation is re-simulated by the twin.
- **Honest AI evaluation:** every model is compared with a simple baseline using a statistical test and is evaluated on a held-out region of the envelope as well as a random split.

- **Anomalies defined by physics:** the detector looks for diagnostics that disagree with the twin, catching faults that a range check cannot see.
- **Local-laboratory defaults:** probe dimensions, sweep settings and chamber geometry follow the COMSATS Plasma Research Laboratory, so students can rehearse the real experiment.

11. Relevance to Course Modules

Table 11.1 shows how the project applies the BS Computer Science course modules.

Table 11.1: Relevance of the project to course modules

Course Module	Application in This Project
Software Engineering / Requirements Engineering	Supervisor questionnaires, mockup-based requirements, event–response tables, traceability, modular design and SDLC management.
Human-Computer Interaction	User classes, clickable prototype, usability and accessibility requirements, guided workflows.
Database Systems	SQLite schema for sessions, imported-file metadata and validation results; transactions and parameterised queries.
Numerical Computing	Root finding for the particle balance, ODE integration of species kinetics, numerical differentiation of probe curves, convolution of spectra.
Calculus, Linear Algebra and Differential Equations	Rate-coefficient integrals over the EEDF, balance equations, 1D diffusion profile.
Probability and Statistics	Least-squares fits with uncertainty, noise models, tolerance-based validation, prediction intervals.
Applied Physics	Plasma physics, atomic spectroscopy and electrostatics behind every model.
Artificial Intelligence / Machine Learning (Core Module 2)	Tree-ensemble regression, Isolation Forest anomaly detection, conformal uncertainty, SHAP explainability, surrogate modelling and optimisation-based inverse design.
Software Quality Engineering	pytest unit and regression tests, continuous integration, measurable non-functional requirements.

12. Tools and Technologies

Table 12.1 lists the tools. Exact versions are not fixed here: they are recorded with pip freeze in requirements.txt when the environment is first created, because library versions change quickly.

Table 12.1: Tools and technologies

Category	Tool / Technology	Version	Rationale
Programming language	Python	3.12+	Scientific ecosystem; current XGBoost and SHAP require Python ≥ 3.12.
User interface	Streamlit	Pinned at setup	Interactive scientific dashboards in pure Python, served locally.
Visualisation	Plotly	Pinned at setup	Interactive spectra, I-V curves, profiles and survival curves.
Numerical computing	NumPy, SciPy	Pinned at setup	Arrays; root finding (brentq); ODE integration; curve fitting; differential evolution; statistical tests.

Category	Tool / Technology	Version	Rationale
Data handling	pandas	Pinned at setup	Parsing of measured .txt/.csv files and table export.
Database	SQLite (sqlite3)	Bundled with Python	Serverless single-file database for sessions and validation records.
Reference data	NIST ASD, LXCat	Snapshot date recorded	Authoritative, citable atomic and cross-section data, installed locally.
Testing	pytest	Pinned at setup	One test file per module; regression grid tests for the solver.
Continuous integration	GitHub Actions (optional)	—	Runs the test suite on every push; seeds fixed for reproducibility.
Version control	Git	2.44+	One commit per working sub-module, in this project's own repository.
AI module	scikit-learn, XGBoost, MAPIE, SHAP	Pinned at setup	Tree-ensemble models, Isolation Forest, conformal intervals and SHAP explanations.
Experiment tracking	MLflow (local file store)	Pinned at setup	Logs every AI training run with its data version, seed and metrics [32].
Prototyping	HTML, CSS, JavaScript	—	Clickable mockup prototype for requirement walkthroughs.
Documentation	Microsoft Word, python-docx	—	Scope document, SRS and report generated in the department format.

13. Project Stakeholders and Roles

Table 13.1 lists the project stakeholders and their responsibilities.

Table 13.1: Project stakeholders and roles

Stakeholder	Role	Responsibilities
Muhammad Rayyan Taimoor Khan (FA23-BCS-072)	Lead developer — digital twin and physics core	Core Module 1 (Stage 1 global model, EEDF and rates, virtual OES, Langmuir probe, Stage 2 model) and Core Module 2 (AI datasets, inference, anomaly detection, surrogate, recommendations); physics and AI tests.
Sadeem Ur Rehman (FA23-BCS-127)	Lead developer — dashboard and analytics	Streamlit screens including the AI screens (M12–M16), Plotly and SHAP visualisation, SQLite sessions and model registry views, measured-data import and validation screens, export and reports, UI/UX.
Dr. Majid Iqbal (Supervisor)	Academic supervisor and domain expert	Defines the experiments and operating envelopes, answers domain questions, supplies sample measured data, reviews milestones.
Students and researchers	Primary end users	Use the platform to rehearse experiments, interpret diagnostics and validate measurements.
FYP evaluation committee	Assessment panel	Assess originality, technical depth, implementation quality, documentation and demonstration.

14. Module-Based Work Division

Table 14.1 assigns primary and supporting ownership for each module.

Table 14.1: Module-based work division

Module / Component	Rayyan (072)	Sadeem (127)	Status
Phase 0: repository, environment, SQLite schema, pytest set-up	Both	Both	Planned
Core 1.1 Low-pressure plasma model (Stage 1)	Primary	Review	Planned
Core 1.2 Virtual optical emission spectroscopy	Primary	Support	Planned
Core 1.3 Virtual Langmuir probe and comparison (Stage 3)	Primary	Support	Planned
Core 1.4 Underwater plasma disinfection model (Stage 2)	Primary	Review	Planned
Core 1.5 Validation and measured data	Support	Primary	Planned
Core 2.1 Spectrum-to-parameter inference	Primary	Support	Planned
Core 2.2 Physics-relationship anomaly detection	Primary	Support	Planned
Core 2.3 Surrogate model and explainability	Primary	Support	Planned
Core 2.4 Counterfactual recommendations	Primary	Support	Planned
Core 2.5 AI training, evaluation and registry	Primary	Support	Planned
S.1 Home, configuration and sessions	Support	Primary	Planned
S.2 Visualisation, reporting and export (incl. AI screens)	Support	Primary	Planned
Testing and integration	Both	Both	Ongoing
Documentation, report and viva preparation	Both	Both	Ongoing

15. Work Breakdown Structure and Gantt Chart

Table 15.1 breaks the work into deliverable tasks, and Table 15.2 places them on a proposed timeline. The final submission deadline has not yet been confirmed, so the timeline is a plan to be agreed with the supervisor. Each module is built and tested on its own before the next one starts, and is committed with the module named in the commit message.

Table 15.1: Work breakdown structure

WBS	Work package	Main tasks	Deliverable
1	Requirements	Supervisor questionnaires; scope document; SRS with mockups	Scope v3.1, SRS v1.1
2	Set-up (Phase 0)	Repository, environment, pinned requirements, SQLite schema, pytest and CI	Running empty app with tests
3	Stage 1 twin (Phases 1–3)	Reference data, global model, EEDF/rates, gas temperature, mixtures, 1D profile, sweeps	Tested Stage 1 engine
4	Virtual OES (Phase 4)	CR model, spectrum synthesis, Boltzmann plot, gas temperature, line ratio	Tested OES module
5	Langmuir probe (Phase 5)	Synthetic I-V, OML analysis, EEPF, comparison	Tested probe module
6	Stage 1 screens (Phase 6)	Home, configuration, results, OES, probe, comparison, sessions	Usable Stage 1 application

WBS	Work package	Main tasks	Deliverable
7	Stage 2 twin (Phase 7)	Bubble global model, liquid transfer, Hom model, underwater OES	Tested Stage 2 module and screens
8	Validation (Phase 8)	Import, metadata, benchmarks, export with citation filter	Validation results within ±30 %
9	AI framework (Phase 9)	Twin datasets, baselines, splits, fault injection, MLflow registry	Locked datasets, registry
10	AI inference (Phase 10)	Spectrum → T_e, n_e, T_{gas} with conformal intervals	Model beating baseline
11	AI surrogate (Phase 11)	Stage 1 and Stage 2 surrogates with SHAP	$R^2 \geq 0.98$ surrogates
12	AI anomalies (Phase 12)	Isolation Forest on physics residuals	Recall ≥ 90 % detector
13	AI recommendations (Phase 13)	Surrogate search verified by the twin	Twin-verified recommendations
14	Testing and report (Phase 14)	Full test pass, final report, presentation and viva	Final submission

Table 15.2: Proposed Gantt chart (October 2026 – June 2027)

WBS	Work package	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
1	Requirements	■	■							
2	Set-up (Phase 0)		■							
3	Stage 1 twin (Phases 1–3)		■	■						
4	Virtual OES (Phase 4)			■	■					
5	Langmuir probe (Phase 5)				■					
6	Stage 1 screens (Phase 6)			■	■					
7	Stage 2 twin (Phase 7)				■	■				
8	Validation (Phase 8)					■				
9	AI framework (Phase 9)				■	■				
10	AI inference (Phase 10)					■	■			
11	AI surrogate (Phase 11)					■	■			
12	AI anomalies (Phase 12)						■	■		
13	AI recommendations (Phase 13)							■	■	
14	Testing and report (Phase 14)								■	■

Shaded cells mark the months in which each work package is active. FYP-I covers October 2026 to January 2027 (Stage 1, its diagnostics and the start of the AI framework on Stage 1 data), and FYP-II covers February to June 2027 (Stage 2, validation, the AI module and the final report).

16. Interface Mockups

The full set of sixteen screen mockups, with the functional requirements derived from each, is given in the SRS [33] and in its clickable HTML prototype. Figure 16.1, Figure 16.2, Figure 16.3 and Figure 16.4 show four representative screens: the Stage 1 chamber configuration, the OES analysis, the Stage 2 underwater configuration and the AI recommendations verified by the twin. Values shown are illustrative placeholders, not results.

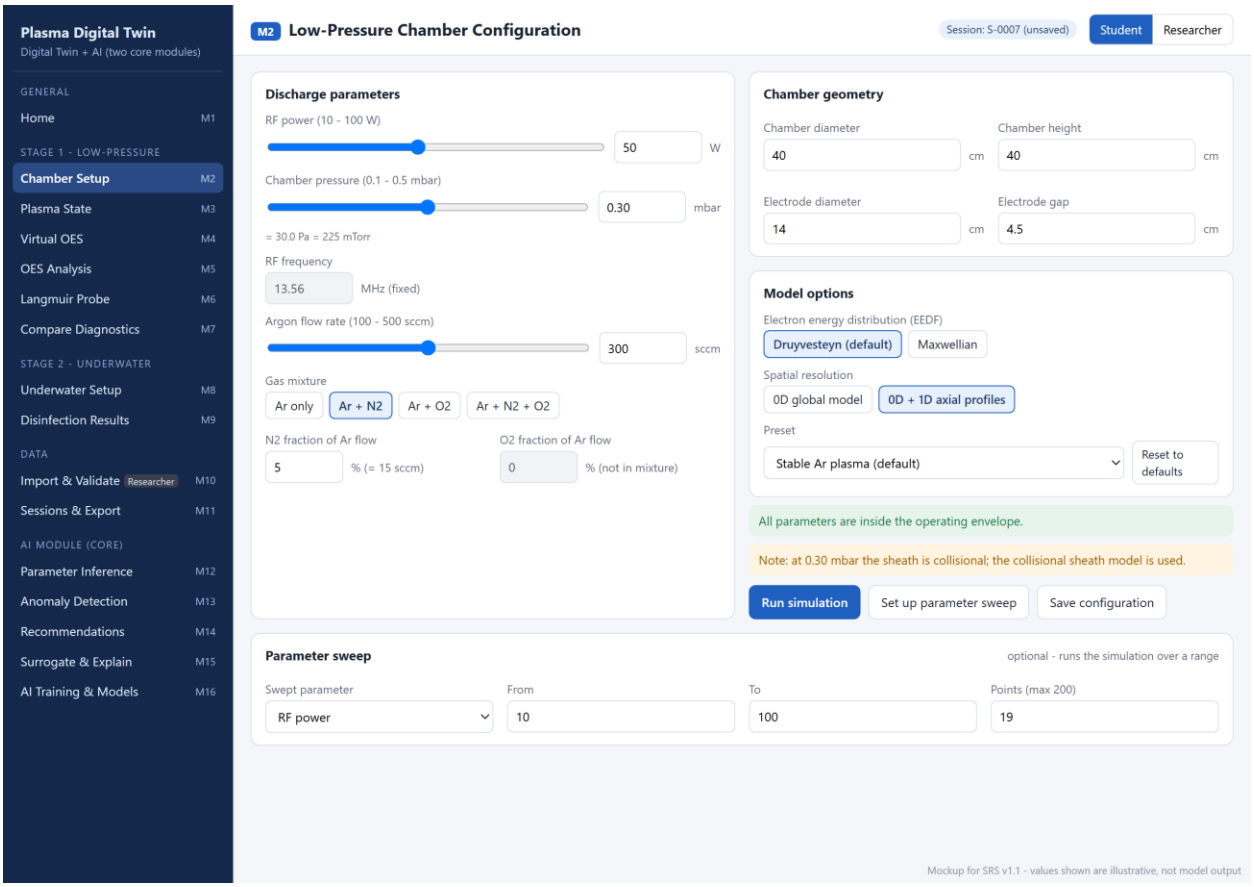


Figure 16.1: Stage 1 low-pressure chamber configuration (SRS Mockup M2)

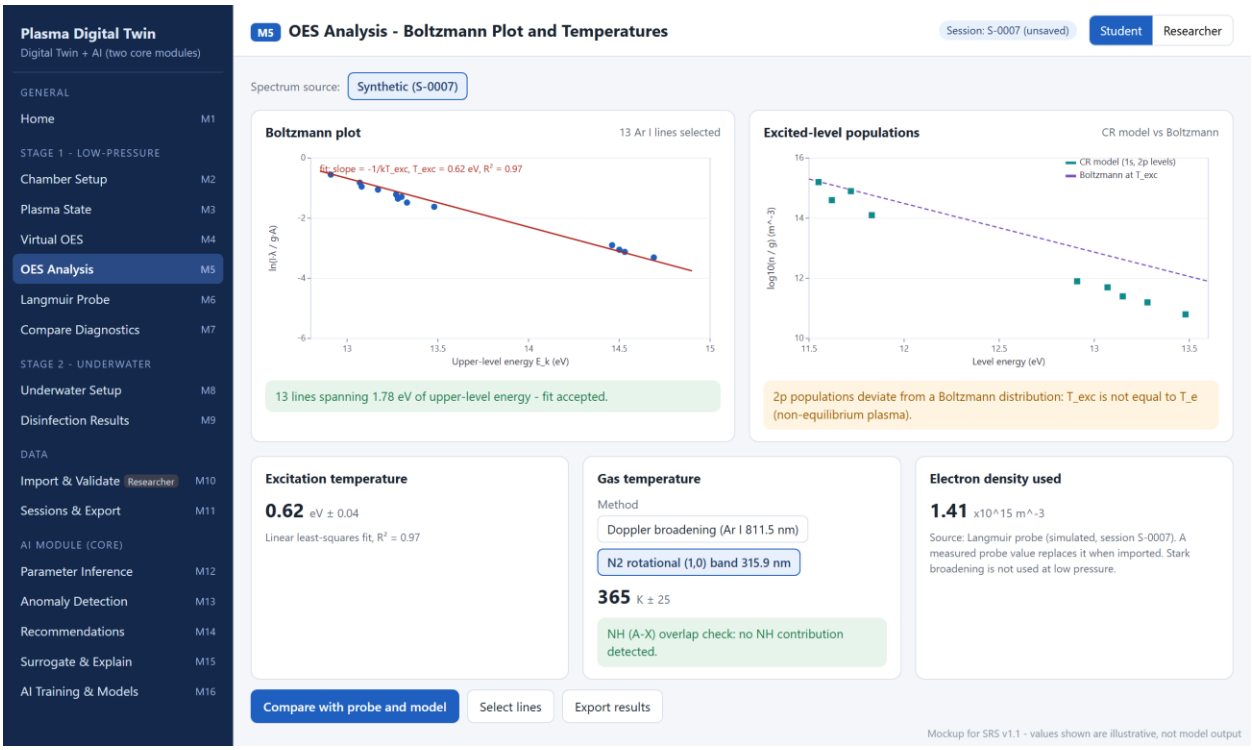


Figure 16.2: OES analysis: Boltzmann plot and temperatures (SRS Mockup M5)

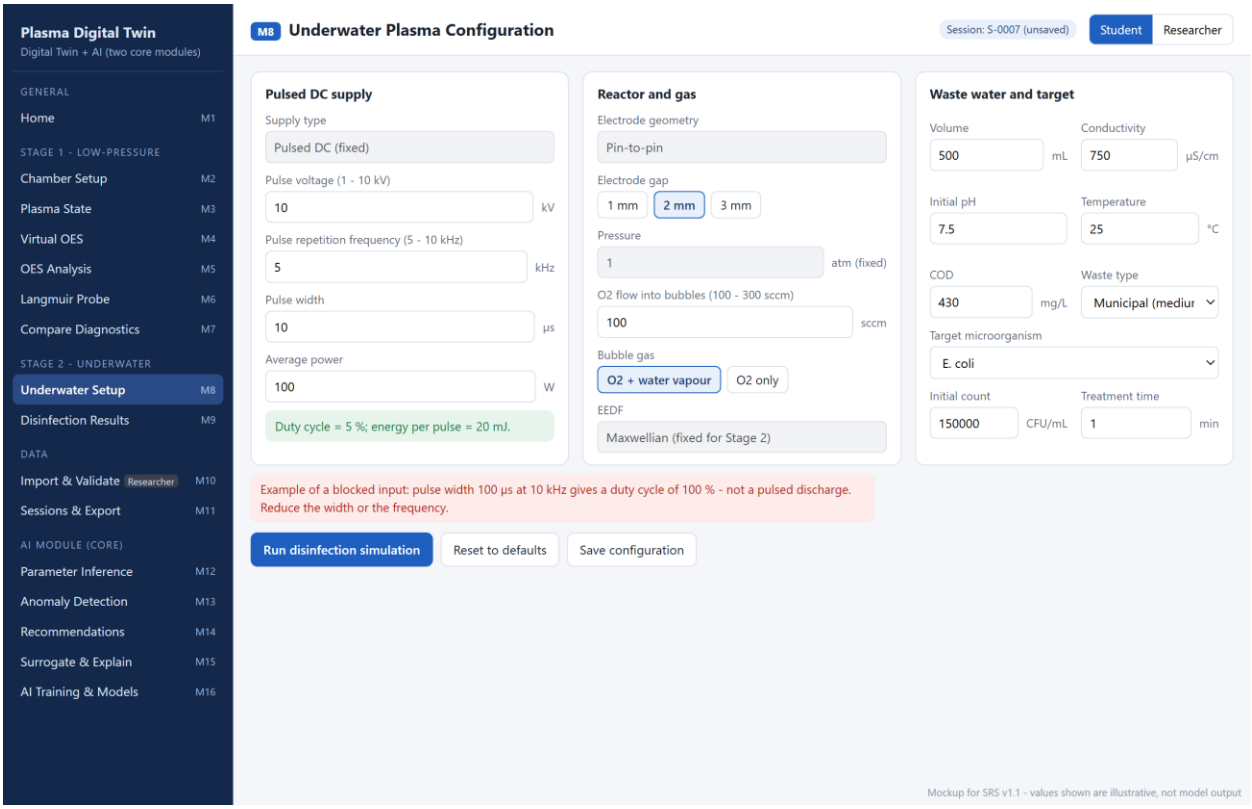


Figure 16.3: Stage 2 underwater plasma configuration (SRS Mockup M8)

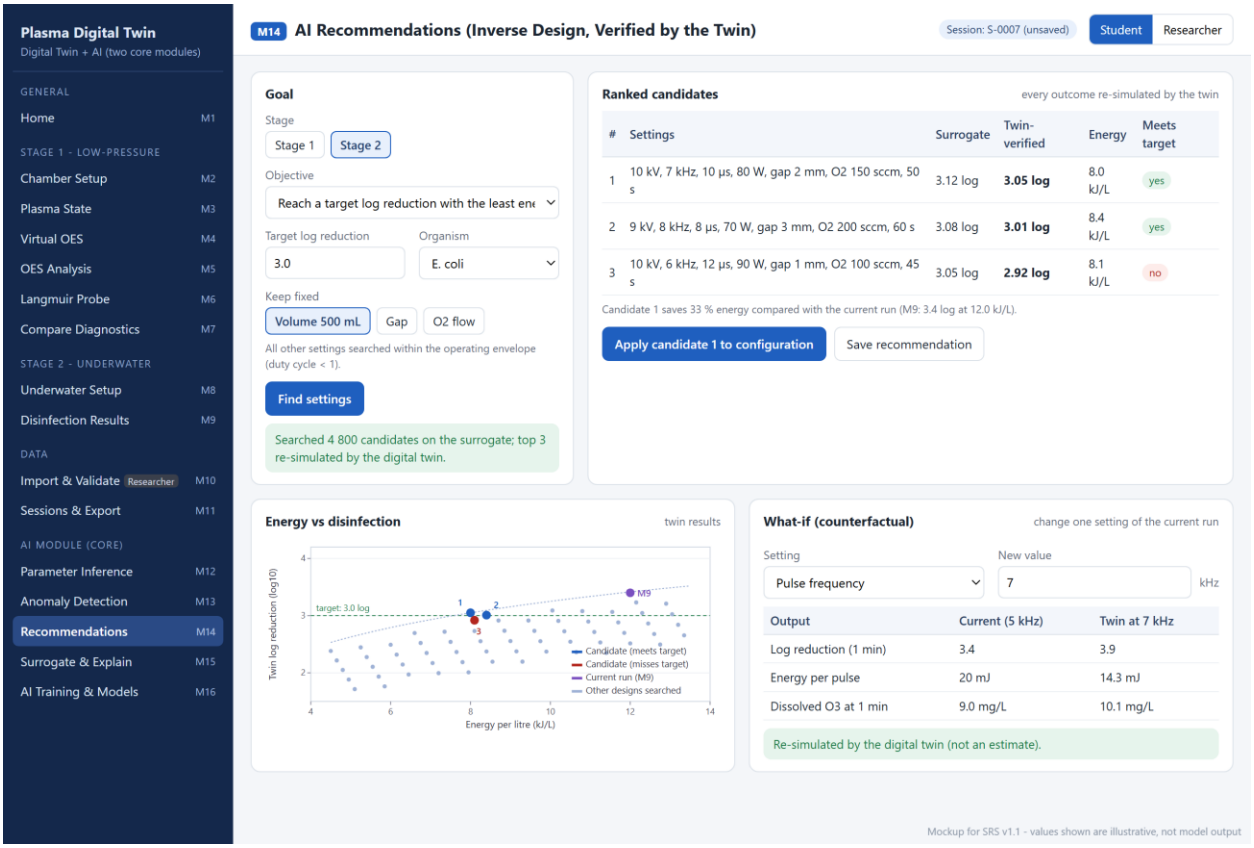


Figure 16.4: AI recommendations verified by the digital twin (SRS Mockup M14)

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18. Plagiarism / Originality Statement

This document has been written by the project team specifically for this final year project. All plasma physics models, data sources and methods are taken from the published literature and data sets cited in Section 17, and the operating parameters come from the supervisor's written answers. No part of this document has been submitted for any other degree or qualification.